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Computational HW#6
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1. Flips coin 7 times, gets 2 heads → $\beta(2,5)$

a. Accuses Hong of biased coin, flips coin: 3 heads, 2 tails (5 flips). Update prior to include new data.

The prior is $\beta(2,5) = P(\theta) = P(H)$

let $\alpha+\beta=\gamma$

$$P(\alpha,\beta,\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{(\alpha-1)}(1-\theta)^{(\beta-1)} = \frac{(\gamma-1)!}{(\alpha-1)!(\beta-1)!}\theta^{(\alpha-1)}(1-\theta)^{(\beta-1)}$$

W/ $\alpha=2, \beta=5, \gamma=7$

$$\Rightarrow \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}\theta^1(1-\theta)^4 = \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}\theta(1-\theta)^4 = P(\theta) = P(H)$$

Likelihood: $P(E,\theta)$, uses new (E) data: $\beta(3,2)$, $n=5$ flips

$$P(h,\theta) = \binom{n}{h}\theta^h(1-\theta)^{(n-h)} = \frac{n!}{h!(n-h)!}\theta^h(1-\theta)^{(n-h)} = P(E,\theta)$$

W/ $n=5, h=3$

$$\Rightarrow \binom{5}{3}\theta^3(1-\theta)^2 = \binom{5}{3}\theta^3(1-\theta)^2$$

Posterior = Likelihood · Prior

$$= \binom{5}{3}\theta^3(1-\theta)^2 \cdot \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}\theta(1-\theta)^4$$

$$= \binom{5}{3}\frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}\theta^4(1-\theta)^6$$

Still need to normalize by $P(E)$, no dependence on θ → integral over all possible $\theta \in [0,1]$

$$P(E) = P(h) = \int_0^1 P(h,\theta)P(\theta)\,d\theta$$

$E = h = 3$ since new data is $h=3$.

$$= \binom{5}{3}\frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}\underbrace{\int_0^1 \theta^4(1-\theta)^6\,d\theta}$$

W/ $\theta^4 \cdot (1-\theta)^6$

Beta integral: $\int_0^1 \theta^{a-1}(1-\theta)^{b-1}\,d\theta = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ $\quad 4=a-1 \quad 6=b-1$, $a=5 \quad b=7$

$$\Rightarrow \binom{5}{3}\frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}\frac{\Gamma(5)\Gamma(7)}{\Gamma(12)}$$

All together:

$$P(H|E) = \frac{\cancel{\binom{5}{3}\frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}}\theta^4(1-\theta)^6}{\cancel{\binom{5}{3}\frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}}\frac{\Gamma(5)\Gamma(7)}{\Gamma(12)}}$$

$$= \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)}\theta^4(1-\theta)^6 = \boxed{\beta(5,7)}$$

Quick Check:
$\beta(2,5)$, data $= \beta(3,2)$
$\beta(2+3, 5+2)$
$\beta(5,7)$ ✓

b. Does 4 more flips and expects to get exactly 2 heads & 2 tails. Calc the binomial coeff that would result if he got exactly 2 heads (Hint: Use beta binomial prediction dist).

$$p(x) = \binom{n}{x} \frac{B(\alpha+x, \beta+n-x)}{B(\alpha,\beta)}, \quad x = 0, 1, \dots, n \qquad n = 4$$

$x = 2?$

$$p(2) = \binom{4}{2} \frac{B(\alpha+2, \beta+4-2)}{B(\alpha,\beta)} \qquad \alpha \text{ & } \beta$$

$$\binom{n}{x} = \frac{n!}{x!(n-x)!} = \frac{4!}{2!2!} = \frac{4\cdot3\cdot\cancel{2}!}{2\cdot\cancel{2}} = \frac{4\cdot3}{2} = \boxed{6 \frac{B(\alpha+2, \beta+2)}{B(\alpha,\beta)}}$$

That's the coefficient

c. Using your eq. from (b) & updated posterior from (a), find the exact probability of getting 2 heads in 4 attempts.

Posterior from a: $B(5,7) = \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)} \theta^4 (\theta-1)^6$

Eq. from b: $6 \frac{B(\alpha+2, \beta+2)}{B(\alpha,\beta)}$ ← Likelihood: beta-binomial dist.

Solving for θ: probability of getting heads

$n=4,\ h=2,\ n-h=2$

Posterior is new prior.

$$P = 6 \frac{B(\alpha+2, \beta+2)}{B(\alpha,\beta)} \qquad \alpha = 5,\ \beta = 7$$

$$\Rightarrow 6 \frac{B(5+2, 7+2)}{B(5,7)} = 6 \frac{B(7,9)}{B(5,7)} = 6 \frac{\Gamma(\cancel{7})\Gamma(9)}{\Gamma(16)} \frac{\Gamma(12)}{\Gamma(5)\Gamma(\cancel{7})} = 6 \frac{\Gamma(9)\Gamma(12)}{\Gamma(16)\Gamma(5)}$$

$\Gamma(n) = (n-1)!$

$\Gamma(9) = 8!$, $\Gamma(12) = 11!$, $\Gamma(16) = 15!$, $\Gamma(5) = 4!$

$$\Rightarrow 6 \frac{(8\cdot7\cdot6\cdot5\cdot\cancel{4}\cdot\cancel{3}\cdot\cancel{2}\cdot\cancel{1})(\cancel{11}\cdot\cancel{10}\cdot\cancel{9}\cdot\cancel{8}\cdot\cancel{7}\cdot\cancel{6}\cdot\cancel{5}\cdot\cancel{4}\cdot\cancel{3}\cdot\cancel{2}\cdot\cancel{1})}{(15\cdot14\cdot13\cdot12\cdot\cancel{11}\cdot\cancel{10}\cdot\cancel{9}\cdot\cancel{8}\cdot\cancel{7}\cdot\cancel{6}\cdot\cancel{5}\cdot\cancel{4}\cdot\cancel{3}\cdot\cancel{2}\cdot\cancel{1})} = 6 \frac{\cancel{8}^{4}\cdot\cancel{7}\cdot\cancel{6}\cdot\cancel{5}}{\cancel{15}_{3}\cdot\cancel{14}_{2}\cdot13\cdot\cancel{12}_{1}} = \boxed{\frac{4}{13} = 0.308 = 30.8\%}$$